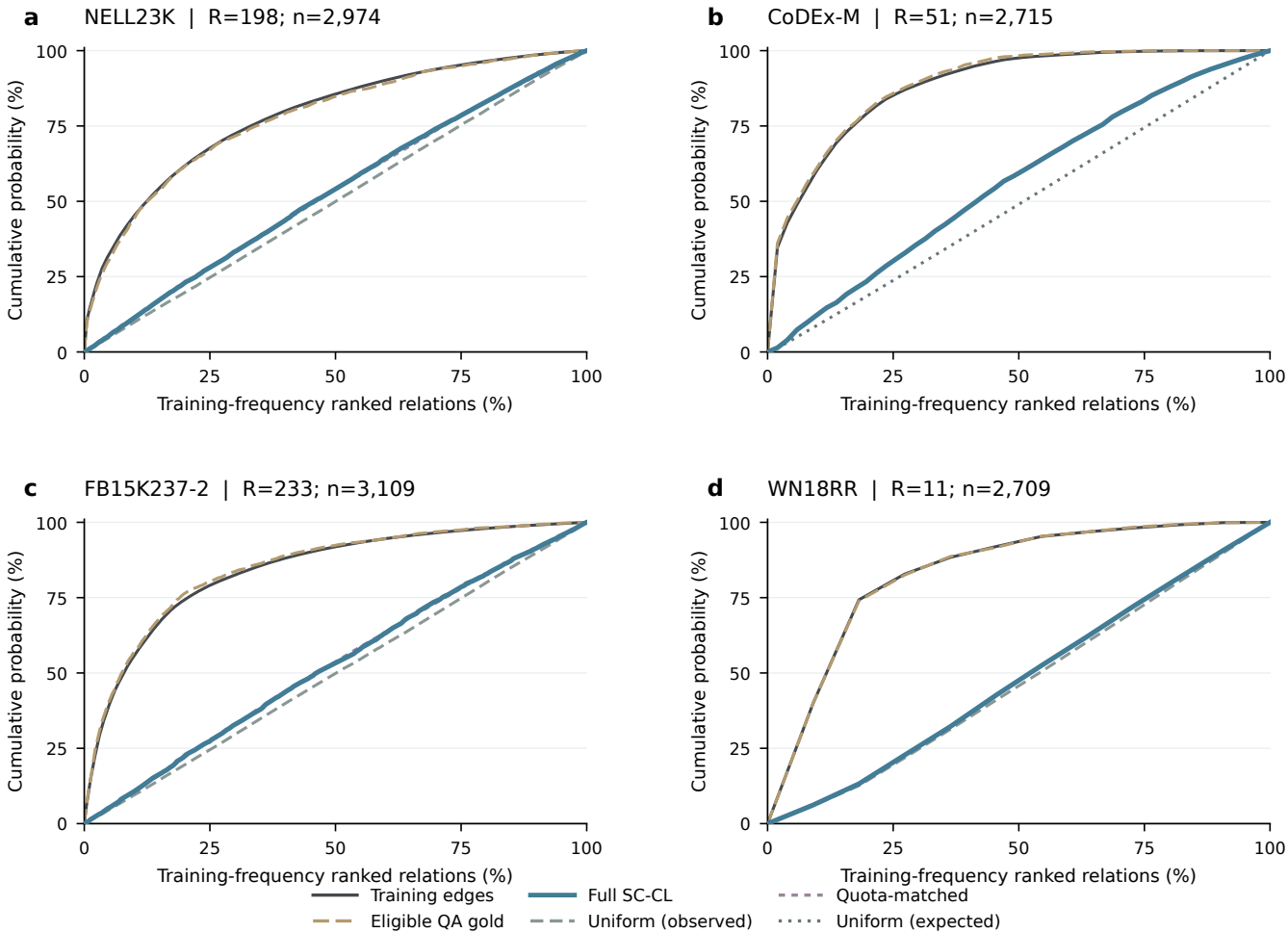
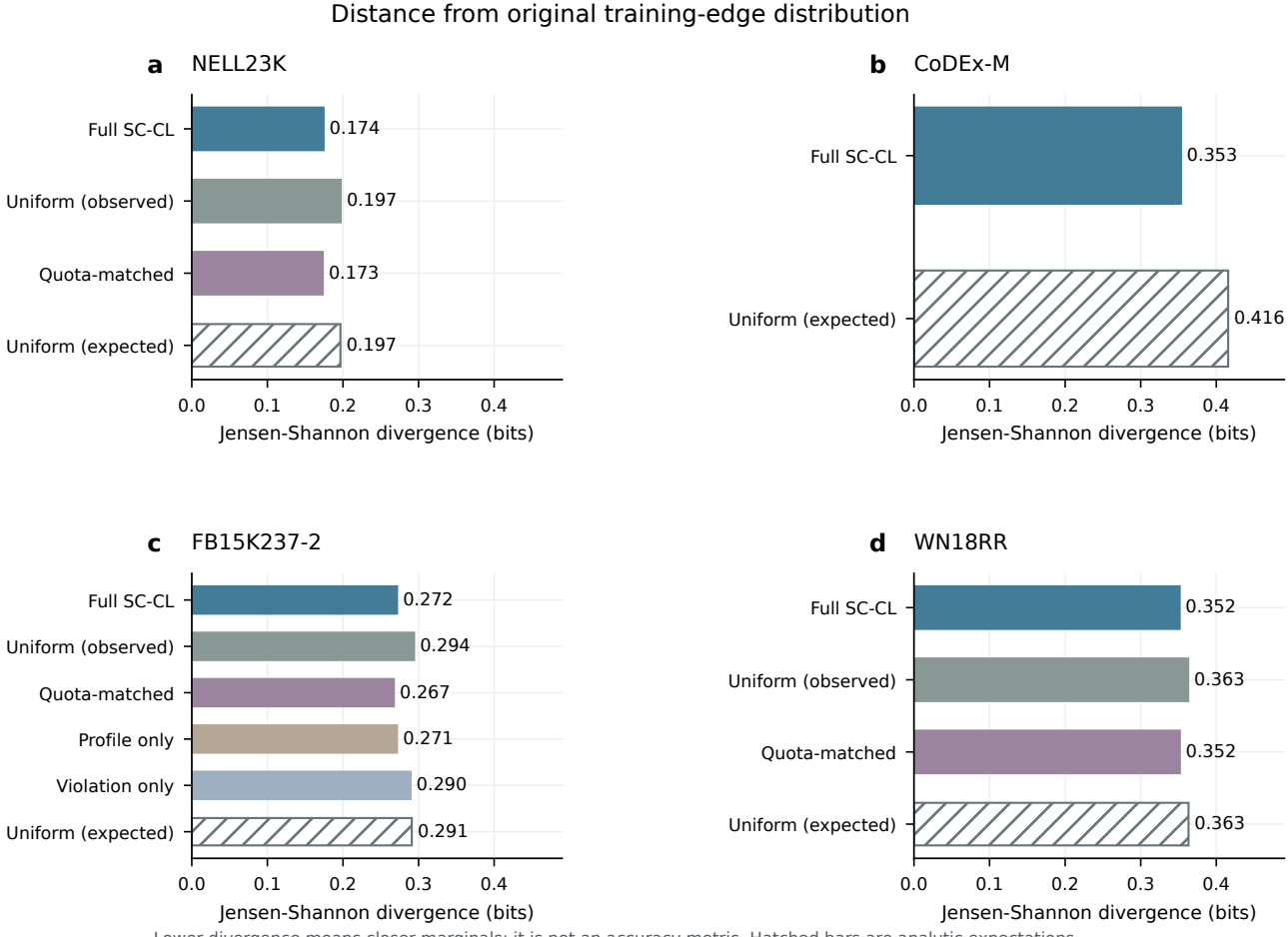
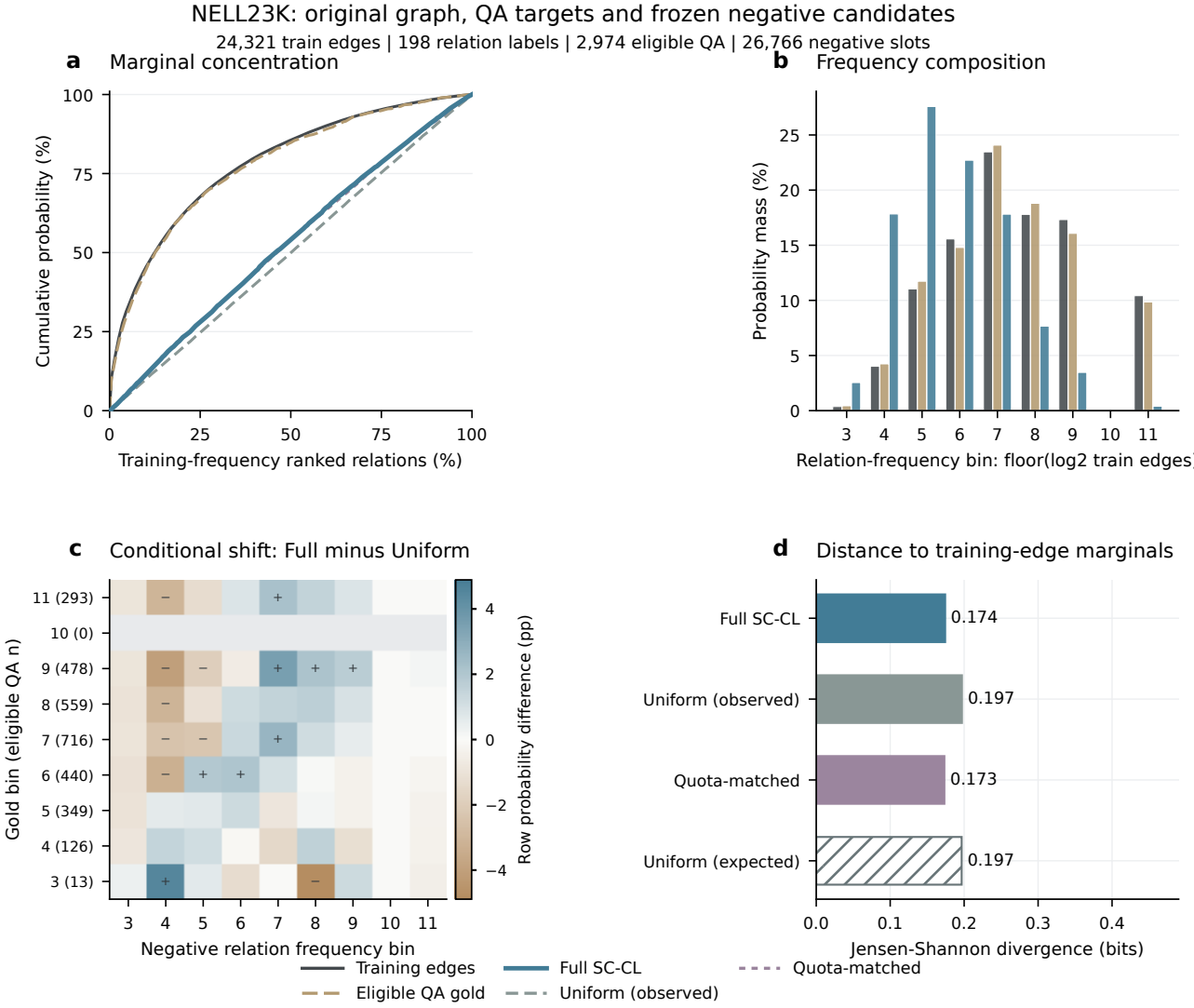
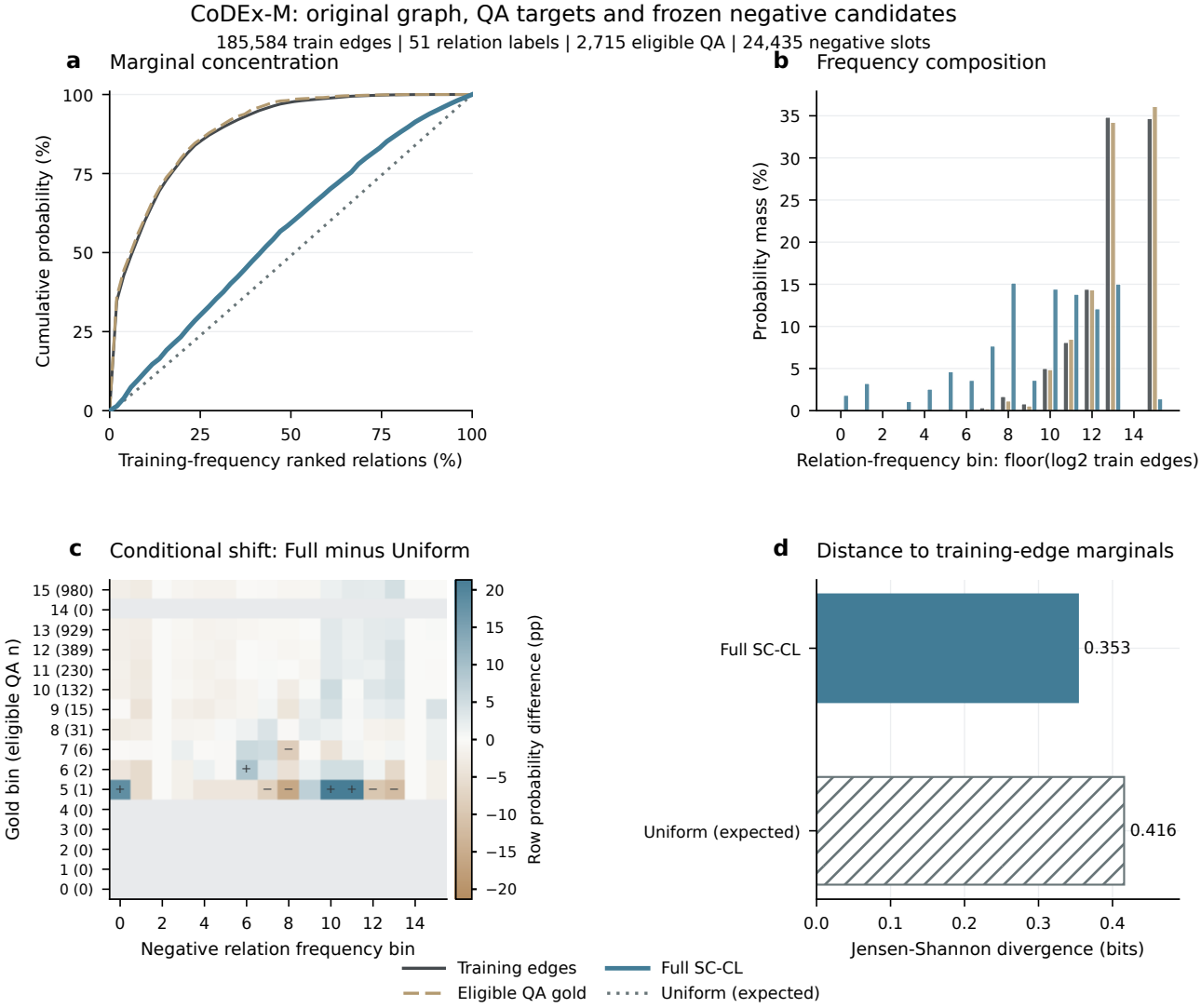
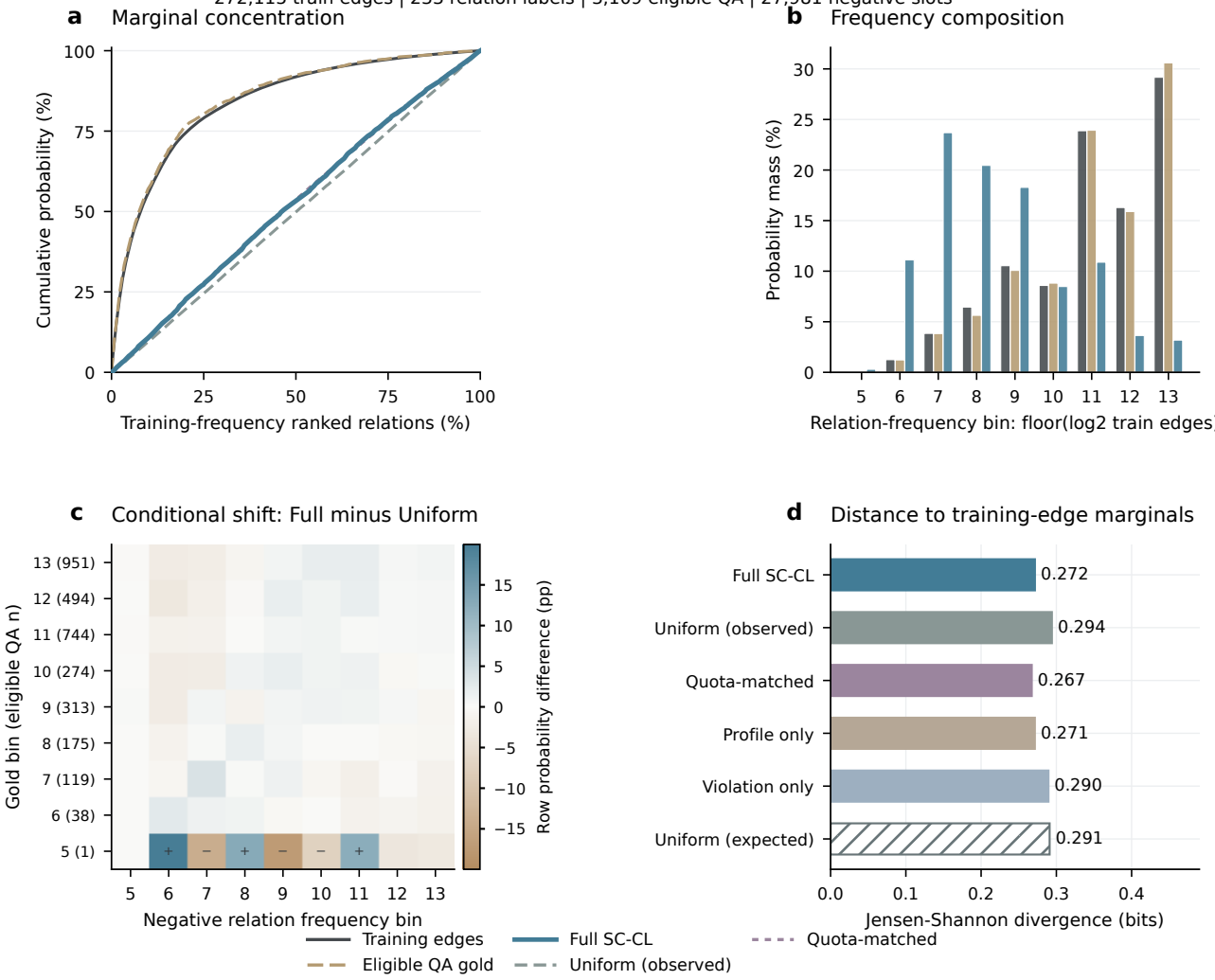

Training-edge, QA-target and negative-sampling distributions
a
NELL23K | R=198; n=2,974
b
CoDEx-M | R=51; n=2,715
c
FB15K237-2 | R=233; n=3,109
d
WN18RR | R=11; n=2,709
Cumulative probability (%)
Training-frequency ranked relations (%)
0
25
50
75
100
Training edges
Eligible QA gold
Full SC-CL
Uniform (observed)
Quota-matched
Uniform (expected)
Relations ranked by training-edge counts. CoDEx-M has no trained controls; its dotted Uniform line is analytic.

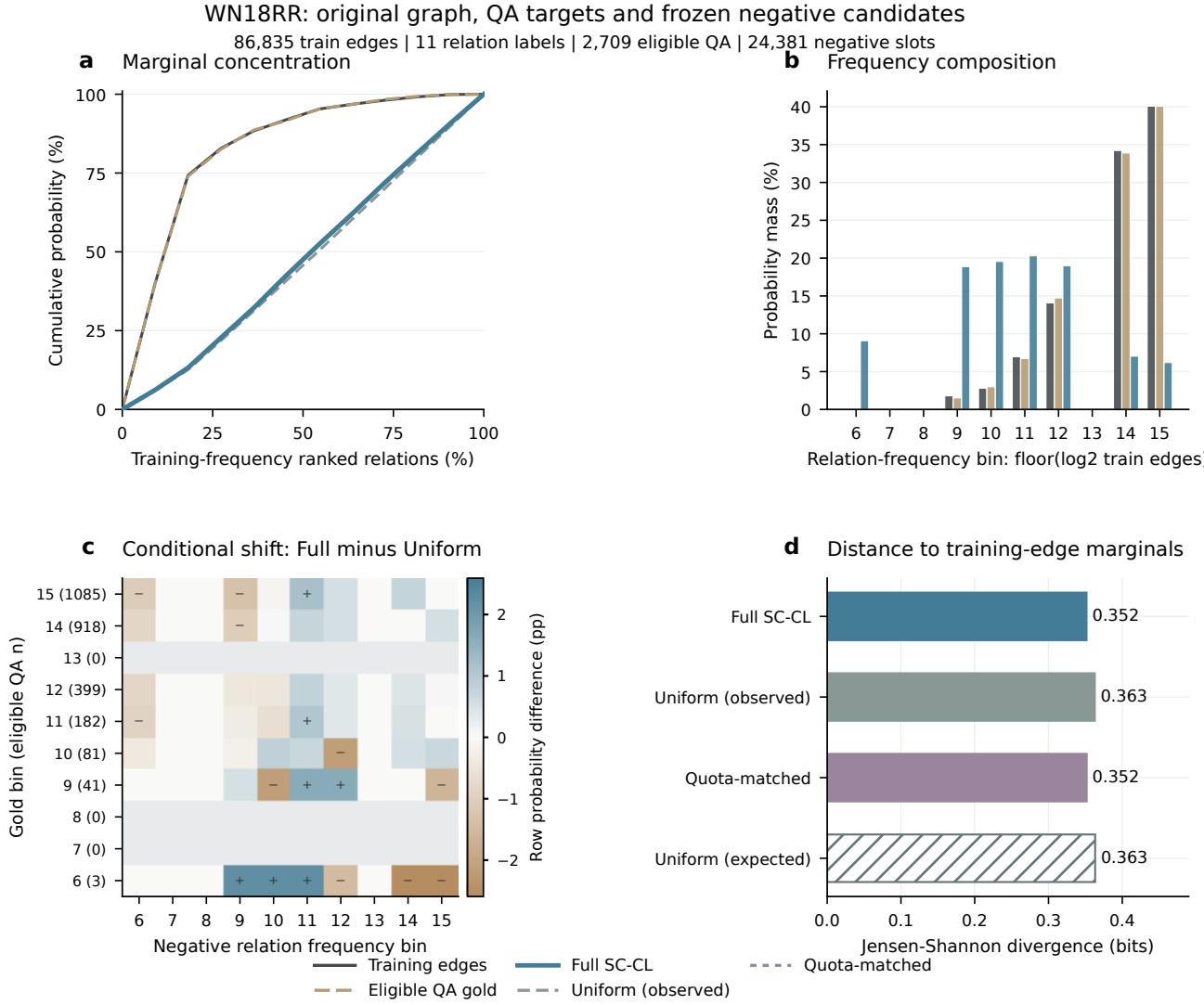

Distance from original training-edge distribution
a
NELL23K
Full SC-CL
Uniform (observed)
Quota-matched
Uniform (expected)
0.174
0.197
0.173
0.197
0.0
0.1
0.2
0.3
0.4
Jensen-Shannon divergence (bits)
b
CoDEx-M
Full SC-CL
Uniform (expected)
0.353
0.416
0.0
0.1
0.2
0.3
0.4
Jensen-Shannon divergence (bits)
c
FB15K237-2
Full SC-CL
Uniform (observed)
Quota-matched
Profile only
Violation only
Uniform (expected)
0.272
0.294
0.267
0.271
0.290
0.291
0.0
0.1
0.2
0.3
0.4
Jensen-Shannon divergence (bits)
d
WN18RR
Full SC-CL
Uniform (observed)
Quota-matched
Uniform (expected)
0.352
0.363
0.352
0.363
0.0
0.1
0.2
0.3
0.4
Jensen-Shannon divergence (bits)
Lower divergence means closer marginals; it is not an accuracy metric. Hatched bars are analytic expectations.


NELL23K: original graph, QA targets and frozen negative candidates
24,321 train edges | 198 relation labels | 2,974 eligible QA | 26,766 negative slots
a
Marginal concentration
Cumulative probability (%)
Training-frequency ranked relations (%)
b
Frequency composition
Probability mass (%)
Relation-frequency bin: floor(log2 train edges)
c
Conditional shift: Full minus Uniform
Gold bin (eligible QA n)
Negative relation frequency bin
Row probability difference (pp)
d
Distance to training-edge marginals
Full SC-CL
Uniform (observed)
Quota-matched
Uniform (expected)
0.174
0.197
0.173
0.197
Jensen-Shannon divergence (bits)
Training edges
Eligible QA gold
Full SC-CL
Uniform (observed)
Quota-matched
Uniform expectation uses each eligible question's legal pool; gray heatmap rows have no observed gold QA.


CoDEx-M: original graph, QA targets and frozen negative candidates
185,584 train edges | 51 relation labels | 2,715 eligible QA | 24,435 negative slots
a
Marginal concentration
Cumulative probability (%)
Training-frequency ranked relations (%)
b
Frequency composition
Probability mass (%)
Relation-frequency bin: floor(log2 train edges)
c
Conditional shift: Full minus Uniform
Gold bin (eligible QA n)
Negative relation frequency bin
Row probability difference (pp)
d
Distance to training-edge marginals
Full SC-CL
0.353
Uniform (expected)
0.416
Jensen-Shannon divergence (bits)
Training edges
Eligible QA gold
Full SC-CL
Uniform (expected)
CoDEx-M: no trained Uniform/quota controls. Dotted lines and hatched bars are analytic legal-pool expectations.



Uniform expectation uses each eligible question's legal pool; gray heatmap rows have no observed gold QA.



WN: 55 historical training pairs unverified; retain parent masks. Conditional baseline is the analytic legal-pool Uniform.